Nokia, Philips, Intel, Schneider Electric, Siemens, Microsoft, Hitachi, Huawei, Ericsson, Toshiba and Oracle have dived headlong into the smart city opportunity. Some companies specialise in a particular area, while others provide end-to-end solutions covering the whole gamut of technologies and systems. There are also innumerable start-ups providing solutions to specific needs of smart cities.

Trip to some of the smart cities

Apart from some of the districts in USA and UK, Aarhus, Amsterdam, Cairo, Lyon, Malaga, Malta, Verona and the Songdo International Business District near Seoul are among regions that have become appreciably smart in many ways in recent years.

Reducing pollution through better traffic management. Las Vegas uses its smart city infrastructure to manage traffic, environmental pollution and other issues. Let's take the example of its traffic signals: If your car is waiting at a signal and there are no approaching vehicles in sight, the signal turns green rather than making your car wait and exhaust more fumes.

Singapore has been implementing different solutions for traffic management since 2014. One among them is a GPS-powered solution that empowers citizens with traffic and roadwork information collected from surveillance cameras installed on roads and taxis. The system also has features like traffic news, travel time calculator, road maps, street directions and parking information. Surveillance cameras alert authorities and vehicle recovery services to road incidents.

Understanding the cause of accidents. On a particular roadway in Jaipur, there were about 4000 accidents annually. The city authorities installed IoT sensors and video cameras to discover the reason. They found that 70 per cent of the accidents happened because drivers went down the road the wrong way. The police used this information to set up sign boards and take other steps to significantly reduce the number of accidents. Sensors in the city also provide information on pollution, availability of parking spots, etc.

Supplementing the city's energy resources. More than 6000 Viennese citizens have invested in community-funded solar and wind power plants in the last five years. Together, these have produced about 25 million kWh of renewable energy, powering almost 15,000 households. Cities across the world are also using smart grids and meters to save power and augment user-contributed power like solar power. Smart meters help users to study their consumption patterns and avail offers like reduced rates for consumption during off-peak hours. While this helps users to save money, the utility is able to conserve power.

IoT infrastructure and mobile apps will let you identify and book parking spots in a smart city (Source: GetMyParking)

Conserving another precious resource—water. Many regions like Berkeley County and Fountain Valley in USA have installed smart water networks to meet water conservation goals. Beyond just remote meter readings, these systems comprising a FlexNet communication system and residential and commercial meters are used to study and understand usage patterns, detect and prevent leakage, and empower consumers to optimise their usage.

The water utility in Little Egg Harbour, a small town in New Jersey, uses technology in a different way. Residents of this town leave during winter and return only when the weather is warm enough. During these cold months, however, the water freezes and pipes tend to break. Using technology from Sensus, the community's municipal utilities authority can monitor residents' homes and quickly respond to problems. This earns them their customers' trust.

Some cities have also started using solutions like WeatherTRAK to optimise their water usage for landscape irrigation. This system uses an IoT machine-to-machine solution and sensors to assess atmospheric and geologic factors and supply just the needed volumes of water instead of constantly dripping water.

Santander in Spain has deployed IoT tracking with a smartphone app, which enables residents to view real-time data on water quality and consumption, track trends over time and receive service alerts. The IoT tracking system provides information about water demand, supply, pressure, quality and other environmental factors, enabling efficient water supply and conservation and sustainable management.

Getting ready for smart waste collection. The Spanish city of Granada is in the process of connecting 14,000 waste bins across the city using sensors. The data collected will be used to identify the bins that need to be emptied, and to optimise pick-up truck routes accordingly. The IoT City Digital Platform in Denmark also includes intelligent waste monitoring using sensors fitted by SmartBin. The Smart City Framework developed by the Sunshine Coast Council in Australia also includes waste management using Enevo's smart fill sensors.

Making sewers smarter too. Cincinnati, USA, has a smart sewer that reduces the overflow of sewage into water bodies. This is a task that normally consumes millions of dollars of the taxpayers' money! The smart sewer system responds to peaks and